Machine learning

With Colab you can import an image dataset, train an image classifier on it, and evaluate the model, all in just a few lines of code.

↳ *1 cell hidden*

More Resources

Working with Notebooks in Colab

- Overview of Colab
- Guide to Markdown
- Importing libraries and installing dependencies
- Saving and loading notebooks in GitHub
- Interactive forms
- Interactive widgets

Working with Data

- Loading data: Drive, Sheets, and Google Cloud Storage
- Charts: visualizing data
- Getting started with BigQuery

Machine Learning Crash Course

These are a few of the notebooks from Google's online Machine Learning course. See the full course website for more.

- Intro to Pandas DataFrame
- Intro to RAPIDS cuDF to accelerate pandas
- Linear regression with tf.keras using synthetic data

Using Accelerated Hardware

- TensorFlow with GPUs
- TensorFlow with TPUs

Featured examples

- Retraining an Image Classifier: Build a Keras model on top of a pre-trained image classifier to distinguish flowers.
- Text Classification: Classify IMDB movie reviews as either *positive* or *negative*.
- Style Transfer: Use deep learning to transfer style between images.
- Multilingual Universal Sentence Encoder Q&A: Use a machine learning model to answer questions from the SQuAD dataset.
- Video Interpolation: Predict what happened in a video between the first and the last frame.

```
import numpy as np
arr = np.array([1, 2,3, 4, 5])
x=arr.copy()
arr[0]=42
print(arr)
print(x)
```

```
[42  2  3  4  5]
[1 2 3 4 5]
```

```
import numpy as np
arr = np.array([1, 2,3, 4, 5, 6, 7, 8, 9, 10, 11, 12])
newarr= arr.reshape(2, 3, 2)
print(newarr)
```

```
[[[ 1  2]
  [ 3  4]
  [ 5  6]]

 [[ 7  8]
  [ 9 10]
  [11 12]]]
```

```
import numpy as np
arr = np.array([[1, 2,3], [4, 5, 6]])
newarr= arr.reshape(-1)
print(newarr)
```

```
[1 2 3 4 5 6]
```

```
import numpy as np
np.identity(2)
```

```
array([[1., 0.],
       [0., 1.]])
```

```
import numpy as np
np.identity(3, dtype=int)
```

```
array([[1, 0, 0],
       [0, 1, 0],
       [0, 0, 1]])
```

```
import numpy as np
np.eye(2)
```

```
array([[1., 0.],
       [0., 1.]])
```

```
import numpy as np
np.eye(3,3)
```

```
array([[1., 0., 0.],
       [0., 1., 0.],
       [0., 0., 1.]])
```

```
import numpy as np
np.eye(2, dtype=int)
```

```
array([[1, 0],
       [0, 1]])
```

```
import numpy as np
np.eye(2,2, dtype=int)
```

```
array([[1, 0],
       [0, 1]])
```

```
import numpy as np
nums=np.eye(5, k=1, dtype=int)
print (nums)
```

```
[[0 1 0 0 0]
 [0 0 1 0 0]
 [0 0 0 1 0]
 [0 0 0 0 1]
 [0 0 0 0 0]]
```